# Standard Normal Distribution Function Table, $\Phi(z)$

| z | -0.09 | -0.08 | -0.07 | -0.06 | -0.05 | -0.04 | -0.03 | -0.02 | -0.01 | 0.00 |
|---|---|---|---|---|---|---|---|---|---|---|
| **-4.0** | 0.00002 | 0.00002 | 0.00002 | 0.00002 | 0.00003 | 0.00003 | 0.00003 | 0.00003 | 0.00003 | 0.00003 |
| **-3.9** | 0.00003 | 0.00003 | 0.00004 | 0.00004 | 0.00004 | 0.00004 | 0.00004 | 0.00004 | 0.00005 | 0.00005 |
| **-3.8** | 0.00005 | 0.00005 | 0.00005 | 0.00006 | 0.00006 | 0.00006 | 0.00006 | 0.00007 | 0.00007 | 0.00007 |
| **-3.7** | 0.00008 | 0.00008 | 0.00008 | 0.00008 | 0.00009 | 0.00009 | 0.00010 | 0.00010 | 0.00010 | 0.00011 |
| **-3.6** | 0.00011 | 0.00012 | 0.00012 | 0.00013 | 0.00013 | 0.00014 | 0.00014 | 0.00015 | 0.00015 | 0.00016 |
| **-3.5** | 0.00017 | 0.00017 | 0.00018 | 0.00019 | 0.00019 | 0.00020 | 0.00021 | 0.00022 | 0.00022 | 0.00023 |
| **-3.4** | 0.00024 | 0.00025 | 0.00026 | 0.00027 | 0.00028 | 0.00029 | 0.00030 | 0.00031 | 0.00032 | 0.00034 |
| **-3.3** | 0.00035 | 0.00036 | 0.00038 | 0.00039 | 0.00040 | 0.00042 | 0.00043 | 0.00045 | 0.00047 | 0.00048 |
| **-3.2** | 0.00050 | 0.00052 | 0.00054 | 0.00056 | 0.00058 | 0.00060 | 0.00062 | 0.00064 | 0.00066 | 0.00069 |
| **-3.1** | 0.00071 | 0.00074 | 0.00076 | 0.00079 | 0.00082 | 0.00084 | 0.00087 | 0.00090 | 0.00094 | 0.00097 |
| **-3.0** | 0.00100 | 0.00104 | 0.00107 | 0.00111 | 0.00114 | 0.00118 | 0.00122 | 0.00126 | 0.00131 | 0.00135 |
| **-2.9** | 0.00139 | 0.00144 | 0.00149 | 0.00154 | 0.00159 | 0.00164 | 0.00169 | 0.00175 | 0.00181 | 0.00187 |
| **-2.8** | 0.00193 | 0.00199 | 0.00205 | 0.00212 | 0.00219 | 0.00226 | 0.00233 | 0.00240 | 0.00248 | 0.00256 |
| **-2.7** | 0.00264 | 0.00272 | 0.00280 | 0.00289 | 0.00298 | 0.00307 | 0.00317 | 0.00326 | 0.00336 | 0.00347 |
| **-2.6** | 0.00357 | 0.00368 | 0.00379 | 0.00391 | 0.00402 | 0.00415 | 0.00427 | 0.00440 | 0.00453 | 0.00466 |
| **-2.5** | 0.00480 | 0.00494 | 0.00508 | 0.00523 | 0.00539 | 0.00554 | 0.00570 | 0.00587 | 0.00604 | 0.00621 |
| **-2.4** | 0.00639 | 0.00657 | 0.00676 | 0.00695 | 0.00714 | 0.00734 | 0.00755 | 0.00776 | 0.00798 | 0.00820 |
| **-2.3** | 0.00842 | 0.00866 | 0.00889 | 0.00914 | 0.00939 | 0.00964 | 0.00990 | 0.01017 | 0.01044 | 0.01072 |
| **-2.2** | 0.01101 | 0.01130 | 0.01160 | 0.01191 | 0.01222 | 0.01255 | 0.01287 | 0.01321 | 0.01355 | 0.01390 |
| **-2.1** | 0.01426 | 0.01463 | 0.01500 | 0.01539 | 0.01578 | 0.01618 | 0.01659 | 0.01700 | 0.01743 | 0.01786 |
| **-2.0** | 0.01831 | 0.01876 | 0.01923 | 0.01970 | 0.02018 | 0.02068 | 0.02118 | 0.02169 | 0.02222 | 0.02275 |
| **-1.9** | 0.02330 | 0.02385 | 0.02442 | 0.02500 | 0.02559 | 0.02619 | 0.02680 | 0.02743 | 0.02807 | 0.02872 |
| **-1.8** | 0.02938 | 0.03005 | 0.03074 | 0.03144 | 0.03216 | 0.03288 | 0.03362 | 0.03438 | 0.03515 | 0.03593 |
| **-1.7** | 0.03673 | 0.03754 | 0.03836 | 0.03920 | 0.04006 | 0.04093 | 0.04182 | 0.04272 | 0.04363 | 0.04457 |
| **-1.6** | 0.04551 | 0.04648 | 0.04746 | 0.04846 | 0.04947 | 0.05050 | 0.05155 | 0.05262 | 0.05370 | 0.05480 |
| **-1.5** | 0.05592 | 0.05705 | 0.05821 | 0.05938 | 0.06057 | 0.06178 | 0.06301 | 0.06426 | 0.06552 | 0.06681 |
| **-1.4** | 0.06811 | 0.06944 | 0.07078 | 0.07215 | 0.07353 | 0.07493 | 0.07636 | 0.07780 | 0.07927 | 0.08076 |
| **-1.3** | 0.08226 | 0.08379 | 0.08534 | 0.08691 | 0.08851 | 0.09012 | 0.09176 | 0.09342 | 0.09510 | 0.09680 |
| **-1.2** | 0.09853 | 0.10027 | 0.10204 | 0.10383 | 0.10565 | 0.10749 | 0.10935 | 0.11123 | 0.11314 | 0.11507 |
| **-1.1** | 0.11702 | 0.11900 | 0.12100 | 0.12302 | 0.12507 | 0.12714 | 0.12924 | 0.13136 | 0.13350 | 0.13567 |
| **-1.0** | 0.13786 | 0.14007 | 0.14231 | 0.14457 | 0.14686 | 0.14917 | 0.15151 | 0.15386 | 0.15625 | 0.15866 |
| **-0.9** | 0.16109 | 0.16354 | 0.16602 | 0.16853 | 0.17106 | 0.17361 | 0.17619 | 0.17879 | 0.18141 | 0.18406 |
| **-0.8** | 0.18673 | 0.18943 | 0.19215 | 0.19489 | 0.19766 | 0.20045 | 0.20327 | 0.20611 | 0.20897 | 0.21186 |
| **-0.7** | 0.21476 | 0.21770 | 0.22065 | 0.22363 | 0.22663 | 0.22965 | 0.23270 | 0.23576 | 0.23885 | 0.24196 |
| **-0.6** | 0.24510 | 0.24825 | 0.25143 | 0.25463 | 0.25785 | 0.26109 | 0.26435 | 0.26763 | 0.27093 | 0.27425 |
| **-0.5** | 0.27760 | 0.28096 | 0.28434 | 0.28774 | 0.29116 | 0.29460 | 0.29806 | 0.30153 | 0.30503 | 0.30854 |
| **-0.4** | 0.31207 | 0.31561 | 0.31918 | 0.32276 | 0.32636 | 0.32997 | 0.33360 | 0.33724 | 0.34090 | 0.34458 |
| **-0.3** | 0.34827 | 0.35197 | 0.35569 | 0.35942 | 0.36317 | 0.36693 | 0.37070 | 0.37448 | 0.37828 | 0.38209 |
| **-0.2** | 0.38591 | 0.38974 | 0.39358 | 0.39743 | 0.40129 | 0.40517 | 0.40905 | 0.41294 | 0.41683 | 0.42074 |
| **-0.1** | 0.42465 | 0.42858 | 0.43251 | 0.43644 | 0.44038 | 0.44433 | 0.44828 | 0.45224 | 0.45620 | 0.46017 |
| **0.0** | 0.46414 | 0.46812 | 0.47210 | 0.47608 | 0.48006 | 0.48405 | 0.48803 | 0.49202 | 0.49601 | 0.50000 |

| z | 0.00 | 0.01 | 0.02 | 0.03 | 0.04 | 0.05 | 0.06 | 0.07 | 0.08 | 0.09 |
|---|---|---|---|---|---|---|---|---|---|---|
| **0.0** | 0.50000 | 0.50399 | 0.50798 | 0.51197 | 0.51595 | 0.51994 | 0.52392 | 0.52790 | 0.53188 | 0.53586 |
| **0.1** | 0.53983 | 0.54380 | 0.54776 | 0.55172 | 0.55567 | 0.55962 | 0.56356 | 0.56749 | 0.57142 | 0.57535 |
| **0.2** | 0.57926 | 0.58317 | 0.58706 | 0.59095 | 0.59483 | 0.59871 | 0.60257 | 0.60642 | 0.61026 | 0.61409 |
| **0.3** | 0.61791 | 0.62172 | 0.62552 | 0.62930 | 0.63307 | 0.63683 | 0.64058 | 0.64431 | 0.64803 | 0.65173 |
| **0.4** | 0.65542 | 0.65910 | 0.66276 | 0.66640 | 0.67003 | 0.67364 | 0.67724 | 0.68082 | 0.68439 | 0.68793 |
| **0.5** | 0.69146 | 0.69497 | 0.69847 | 0.70194 | 0.70540 | 0.70884 | 0.71226 | 0.71566 | 0.71904 | 0.72240 |
| **0.6** | 0.72575 | 0.72907 | 0.73237 | 0.73565 | 0.73891 | 0.74215 | 0.74537 | 0.74857 | 0.75175 | 0.75490 |
| **0.7** | 0.75804 | 0.76115 | 0.76424 | 0.76730 | 0.77035 | 0.77337 | 0.77637 | 0.77935 | 0.78230 | 0.78524 |
| **0.8** | 0.78814 | 0.79103 | 0.79389 | 0.79673 | 0.79955 | 0.80234 | 0.80511 | 0.80785 | 0.81057 | 0.81327 |
| **0.9** | 0.81594 | 0.81859 | 0.82121 | 0.82381 | 0.82639 | 0.82894 | 0.83147 | 0.83398 | 0.83646 | 0.83891 |
| **1.0** | 0.84134 | 0.84375 | 0.84614 | 0.84849 | 0.85083 | 0.85314 | 0.85543 | 0.85769 | 0.85993 | 0.86214 |
| **1.1** | 0.86433 | 0.86650 | 0.86864 | 0.87076 | 0.87286 | 0.87493 | 0.87698 | 0.87900 | 0.88100 | 0.88298 |
| **1.2** | 0.88493 | 0.88686 | 0.88877 | 0.89065 | 0.89251 | 0.89435 | 0.89617 | 0.89796 | 0.89973 | 0.90147 |
| **1.3** | 0.90320 | 0.90490 | 0.90658 | 0.90824 | 0.90988 | 0.91149 | 0.91309 | 0.91466 | 0.91621 | 0.91774 |
| **1.4** | 0.91924 | 0.92073 | 0.92220 | 0.92364 | 0.92507 | 0.92647 | 0.92785 | 0.92922 | 0.93056 | 0.93189 |
| **1.5** | 0.93319 | 0.93448 | 0.93574 | 0.93699 | 0.93822 | 0.93943 | 0.94062 | 0.94179 | 0.94295 | 0.94408 |
| **1.6** | 0.94520 | 0.94630 | 0.94738 | 0.94845 | 0.94950 | 0.95053 | 0.95154 | 0.95254 | 0.95352 | 0.95449 |
| **1.7** | 0.95543 | 0.95637 | 0.95728 | 0.95818 | 0.95907 | 0.95994 | 0.96080 | 0.96164 | 0.96246 | 0.96327 |
| **1.8** | 0.96407 | 0.96485 | 0.96562 | 0.96638 | 0.96712 | 0.96784 | 0.96856 | 0.96926 | 0.96995 | 0.97062 |
| **1.9** | 0.97128 | 0.97193 | 0.97257 | 0.97320 | 0.97381 | 0.97441 | 0.97500 | 0.97558 | 0.97615 | 0.97670 |
| **2.0** | 0.97725 | 0.97778 | 0.97831 | 0.97882 | 0.97932 | 0.97982 | 0.98030 | 0.98077 | 0.98124 | 0.98169 |
| **2.1** | 0.98214 | 0.98257 | 0.98300 | 0.98341 | 0.98382 | 0.98422 | 0.98461 | 0.98500 | 0.98537 | 0.98574 |
| **2.2** | 0.98610 | 0.98645 | 0.98679 | 0.98713 | 0.98745 | 0.98778 | 0.98809 | 0.98840 | 0.98870 | 0.98899 |
| **2.3** | 0.98928 | 0.98956 | 0.98983 | 0.99010 | 0.99036 | 0.99061 | 0.99086 | 0.99111 | 0.99134 | 0.99158 |
| **2.4** | 0.99180 | 0.99202 | 0.99224 | 0.99245 | 0.99266 | 0.99286 | 0.99305 | 0.99324 | 0.99343 | 0.99361 |
| **2.5** | 0.99379 | 0.99396 | 0.99413 | 0.99430 | 0.99446 | 0.99461 | 0.99477 | 0.99492 | 0.99506 | 0.99520 |
| **2.6** | 0.99534 | 0.99547 | 0.99560 | 0.99573 | 0.99585 | 0.99598 | 0.99609 | 0.99621 | 0.99632 | 0.99643 |
| **2.7** | 0.99653 | 0.99664 | 0.99674 | 0.99683 | 0.99693 | 0.99702 | 0.99711 | 0.99720 | 0.99728 | 0.99736 |
| **2.8** | 0.99744 | 0.99752 | 0.99760 | 0.99767 | 0.99774 | 0.99781 | 0.99788 | 0.99795 | 0.99801 | 0.99807 |
| **2.9** | 0.99813 | 0.99819 | 0.99825 | 0.99831 | 0.99836 | 0.99841 | 0.99846 | 0.99851 | 0.99856 | 0.99861 |
| **3.0** | 0.99865 | 0.99869 | 0.99874 | 0.99878 | 0.99882 | 0.99886 | 0.99889 | 0.99893 | 0.99896 | 0.99900 |
| **3.1** | 0.99903 | 0.99906 | 0.99910 | 0.99913 | 0.99916 | 0.99918 | 0.99921 | 0.99924 | 0.99926 | 0.99929 |
| **3.2** | 0.99931 | 0.99934 | 0.99936 | 0.99938 | 0.99940 | 0.99942 | 0.99944 | 0.99946 | 0.99948 | 0.99950 |
| **3.3** | 0.99952 | 0.99953 | 0.99955 | 0.99957 | 0.99958 | 0.99960 | 0.99961 | 0.99962 | 0.99964 | 0.99965 |
| **3.4** | 0.99966 | 0.99968 | 0.99969 | 0.99970 | 0.99971 | 0.99972 | 0.99973 | 0.99974 | 0.99975 | 0.99976 |
| **3.5** | 0.99977 | 0.99978 | 0.99978 | 0.99979 | 0.99980 | 0.99981 | 0.99981 | 0.99982 | 0.99983 | 0.99983 |
| **3.6** | 0.99984 | 0.99985 | 0.99985 | 0.99986 | 0.99986 | 0.99987 | 0.99987 | 0.99988 | 0.99988 | 0.99989 |
| **3.7** | 0.99989 | 0.99990 | 0.99990 | 0.99990 | 0.99991 | 0.99991 | 0.99992 | 0.99992 | 0.99992 | 0.99992 |
| **3.8** | 0.99993 | 0.99993 | 0.99993 | 0.99994 | 0.99994 | 0.99994 | 0.99994 | 0.99995 | 0.99995 | 0.99995 |
| **3.9** | 0.99995 | 0.99995 | 0.99996 | 0.99996 | 0.99996 | 0.99996 | 0.99996 | 0.99996 | 0.99997 | 0.99997 |
| **4.0** | 0.99997 | 0.99997 | 0.99997 | 0.99997 | 0.99997 | 0.99997 | 0.99998 | 0.99998 | 0.99998 | 0.99998 |

# Abridged Normal Table

| z | Φ(z) | z | Φ(z) | z | Φ(z) | z | Φ(z) | z | Φ(z) | z | Φ(z) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 0.99997 | 2.64 | 0.99585 | 1.28 | 0.89973 | -0.08 | 0.46812 | -1.44 | 0.07493 | -2.8 | 0.00256 |
| 3.96 | 0.99996 | 2.60 | 0.99534 | 1.24 | 0.89251 | -0.12 | 0.45224 | -1.48 | 0.06944 | -2.84 | 0.00226 |
| 3.92 | 0.99996 | 2.56 | 0.99477 | 1.20 | 0.88493 | -0.16 | 0.43644 | -1.52 | 0.06426 | -2.88 | 0.00199 |
| 3.88 | 0.99995 | 2.52 | 0.99413 | 1.16 | 0.87698 | -0.20 | 0.42074 | -1.56 | 0.05938 | -2.92 | 0.00175 |
| 3.84 | 0.99994 | 2.48 | 0.99343 | 1.12 | 0.86864 | -0.24 | 0.40517 | -1.60 | 0.05480 | -2.96 | 0.00154 |
| 3.8 | 0.99993 | 2.44 | 0.99266 | 1.08 | 0.85993 | -0.28 | 0.38974 | -1.64 | 0.05050 | -3 | 0.00135 |
| 3.76 | 0.99992 | 2.40 | 0.99180 | 1.04 | 0.85083 | -0.32 | 0.37448 | -1.68 | 0.04648 | -3.04 | 0.00118 |
| 3.72 | 0.99990 | 2.36 | 0.99086 | 1.00 | 0.84134 | -0.36 | 0.35942 | -1.72 | 0.04272 | -3.08 | 0.00104 |
| 3.68 | 0.99988 | 2.32 | 0.98983 | 0.96 | 0.83147 | -0.40 | 0.34458 | -1.76 | 0.03920 | -3.12 | 0.00090 |
| 3.64 | 0.99986 | 2.28 | 0.98870 | 0.92 | 0.82121 | -0.44 | 0.32997 | -1.80 | 0.03593 | -3.16 | 0.00079 |
| 3.6 | 0.99984 | 2.24 | 0.98745 | 0.88 | 0.81057 | -0.48 | 0.31561 | -1.84 | 0.03288 | -3.2 | 0.00069 |
| 3.56 | 0.99981 | 2.20 | 0.98610 | 0.84 | 0.79955 | -0.52 | 0.30153 | -1.88 | 0.03005 | -3.24 | 0.00060 |
| 3.52 | 0.99978 | 2.16 | 0.98461 | 0.80 | 0.78814 | -0.56 | 0.28774 | -1.92 | 0.02743 | -3.28 | 0.00052 |
| 3.48 | 0.99975 | 2.12 | 0.98300 | 0.76 | 0.77637 | -0.60 | 0.27425 | -1.96 | 0.02500 | -3.32 | 0.00045 |
| 3.44 | 0.99971 | 2.08 | 0.98124 | 0.72 | 0.76424 | -0.64 | 0.26109 | -2.00 | 0.02275 | -3.36 | 0.00039 |
| 3.40 | 0.99966 | 2.04 | 0.97932 | 0.68 | 0.75175 | -0.68 | 0.24825 | -2.04 | 0.02068 | -3.4 | 0.00034 |
| 3.36 | 0.99961 | 2.00 | 0.97725 | 0.64 | 0.73891 | -0.72 | 0.23576 | -2.08 | 0.01876 | -3.44 | 0.00029 |
| 3.32 | 0.99955 | 1.96 | 0.97500 | 0.60 | 0.72575 | -0.76 | 0.22363 | -2.12 | 0.01700 | -3.48 | 0.00025 |
| 3.28 | 0.99948 | 1.92 | 0.97257 | 0.56 | 0.71226 | -0.80 | 0.21186 | -2.16 | 0.01539 | -3.52 | 0.00022 |
| 3.24 | 0.99940 | 1.88 | 0.96995 | 0.52 | 0.69847 | -0.84 | 0.20045 | -2.20 | 0.01390 | -3.56 | 0.00019 |
| 3.20 | 0.99931 | 1.84 | 0.96712 | 0.48 | 0.68439 | -0.88 | 0.18943 | -2.24 | 0.01255 | -3.6 | 0.00016 |
| 3.16 | 0.99921 | 1.80 | 0.96407 | 0.44 | 0.67003 | -0.92 | 0.17879 | -2.28 | 0.01130 | -3.64 | 0.00014 |
| 3.12 | 0.99910 | 1.76 | 0.96080 | 0.40 | 0.65542 | -0.96 | 0.16853 | -2.32 | 0.01017 | -3.68 | 0.00012 |
| 3.08 | 0.99896 | 1.72 | 0.95728 | 0.36 | 0.64058 | -1.00 | 0.15866 | -2.36 | 0.00914 | -3.72 | 0.00010 |
| 3.04 | 0.99882 | 1.68 | 0.95352 | 0.32 | 0.62552 | -1.04 | 0.14917 | -2.40 | 0.00820 | -3.76 | 0.00008 |
| 3.00 | 0.99865 | 1.64 | 0.94950 | 0.28 | 0.61026 | -1.08 | 0.14007 | -2.44 | 0.00734 | -3.8 | 0.00007 |
| 2.96 | 0.99846 | 1.60 | 0.94520 | 0.24 | 0.59483 | -1.12 | 0.13136 | -2.48 | 0.00657 | -3.84 | 0.00006 |
| 2.92 | 0.99825 | 1.56 | 0.94062 | 0.20 | 0.57926 | -1.16 | 0.12302 | -2.52 | 0.00587 | -3.88 | 0.00005 |
| 2.88 | 0.99801 | 1.52 | 0.93574 | 0.16 | 0.56356 | -1.20 | 0.11507 | -2.56 | 0.00523 | -3.92 | 0.00004 |
| 2.84 | 0.99774 | 1.48 | 0.93056 | 0.12 | 0.54776 | -1.24 | 0.10749 | -2.6 | 0.00466 | -3.96 | 0.00004 |
| 2.80 | 0.99744 | 1.44 | 0.92507 | 0.08 | 0.53188 | -1.28 | 0.10027 | -2.64 | 0.00415 | -4 | 0.00003 |
| 2.76 | 0.99711 | 1.40 | 0.91924 | 0.04 | 0.51595 | -1.32 | 0.09342 | -2.68 | 0.00368 | | |
| 2.72 | 0.99674 | 1.36 | 0.91309 | 0.00 | 0.50000 | -1.36 | 0.08691 | -2.72 | 0.00326 | | |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 2.68 | 0.99632 | 1.32 | 0.90658 | -0.04 | 0.48405 | -1.40 | 0.08076 | -2.76 | 0.00289 | |